

TABLE OF CONTENT

Chapter 1: Introduction

Background of the Study

The rapid digital transformation in the Philippines has drastically increased the volume of personal data collected, processed, and stored by various organizations. To protect the fundamental human right to privacy, the Philippine government enacted Republic Act No. 10173, widely known as the Data Privacy Act of 2012 (DPA). This law mandates that all public and private entities must protect personal information and uphold the rights of data subjects.

Despite the law's clear directives, many organizations struggle to comply due to reliance on manual tracking methods. Data Protection Officers (DPOs) are often overwhelmed by the administrative burden of monitoring consent, tracking data processing activities across different departments, and preparing compliance reports. This reliance on manual administration increases the risk of human error, data breaches, and penalties for non-compliance from the National Privacy Commission (NPC). Therefore, there is a critical need for an automated Data Privacy Compliance System—a centralized web platform designed to enforce "Privacy by Design," manage consent lifecycles, and streamline regulatory reporting

Statement of the Problem

The primary problem this study addresses is the inefficiency and high risk of non-compliance

associated with manual data privacy management in organizations. Specifically, this study seeks to answer the following:

1. How can the organization effectively track and manage user consent across multiple data collection and processing touchpoints without relying on fragmented manual records?

Organizations collect personal information through various channels such as registration forms, websites, employee records, customer transactions, and other digital services.

When consent records are stored separately or managed manually, it becomes difficult to determine whether valid consent has been obtained, updated, or withdrawn. This study seeks to determine how a centralized Consent Management Module can securely record, monitor, and update the consent status of data subjects while ensuring that organizations maintain accurate documentation and comply with the consent requirements of the Data Privacy Act of 2012.

2. How can the Data Protection Officer (DPO) efficiently monitor, document, and manage all data processing activities to ensure that they comply with the principles of transparency, legitimate purpose, and proportionality under the Data Privacy Act of 2012?

One of the major responsibilities of a Data Protection Officer is ensuring that every activity involving personal information is properly documented and conducted only for authorized and lawful purposes. However, manual monitoring methods often result in incomplete records, inconsistencies, and delays in identifying unauthorized or unnecessary data processing activities. This study aims to determine how a centralized Data Processing Tracker can provide comprehensive logging, monitoring, and verification of all data processing operations, enabling the DPO to improve accountability, maintain transparency, and strengthen compliance with the provisions of the Data Privacy Act.

3. What features and functionalities should be incorporated into the proposed Data Privacy Compliance System to automatically generate accurate compliance reports, audit trails, and documentation required by the National Privacy Commission (NPC), while minimizing manual effort and reducing the risk of human error?

Preparing compliance reports for audits and inspections is often a time-consuming process when organizations depend on manually compiled records from different departments. Inaccurate or incomplete reports may lead to compliance issues and increased administrative workload for the Data Protection Officer. Therefore, this study seeks to identify the essential system features that will automate the generation of compliance reports, audit trails, and other regulatory documents required by the National Privacy Commission. The study also aims to determine how these automated capabilities can improve reporting efficiency, reduce human error, and assist organizations in maintaining continuous compliance with Philippine data privacy regulations.

Objectives of the study

General Objective: To develop and implement a web-based Data Privacy Compliance System that assists organizations in managing the collection, processing, storage, and protection of personal data in accordance with RA 10173.

Specific Objectives:

1. To design a Consent Management Module that records, tracks, and updates the consent status of data subjects securely.
2. To develop a Data Processing Tracker that logs all internal data handling activities, ensuring transparency and accountability.

3. To create an Automated Reporting Dashboard that generates real-time compliance reports and audit trails for the DPO and NPC inspections.
4. To evaluate the system's effectiveness and usability based on standard software quality metrics.

Scope and Limitations

Scope: The project focuses on developing a web-based compliance management system tailored for the organization's Data Protection Officer and administrative staff. The system covers consent lifecycle management, data processing activity logging, and the generation of compliance reports. It incorporates basic role-based access control (RBAC) to ensure only authorized personnel can view sensitive compliance logs.

Limitations: The system is designed as a management and tracking portal; it does not automatically scan external databases or physical documents for personally identifiable information (PII). It relies on the accurate input and logging of data processing activities by the organization's staff. Furthermore, the system is explicitly mapped to the Philippine Data Privacy Act of 2012 and may not inherently cover foreign regulations like the European Union's GDPR without further modification.

Significance of the Study

- **For the Organization:** Provides a centralized, automated tool to mitigate legal risks, avoid hefty NPC fines, and build client trust by demonstrating a commitment to data protection.
- **For the Data Protection Officer (DPO):** Significantly reduces the administrative workload by automating consent tracking and report generation, allowing the DPO to focus on security strategy.
- **For the Data Subjects:** Ensures their personal information is handled legally and transparently, and that their rights (such as the right to be informed and the right to

erasure) are respected.

- **For Future Researchers:** Serves as a foundational reference for developing locally compliant privacy-enhancing technologies (PETs) within the Philippine regulatory framework.

Chapter 2: Review of Related Literature and Studies

This chapter presents a comprehensive review of related literature, existing systems, and relevant laws that serve as the foundation for developing the Data Privacy Compliance System.

Related Literatures

Foreign Literatures

1. **Betkier, M. (2019). Moving beyond consent in data privacy law: An effective privacy management system for internet services.**

This doctoral thesis explores the limitations of relying solely on user consent in digital environments. Betkier argues that users often experience "consent fatigue," rendering traditional tick-box agreements ineffective. The study is highly relevant to the proposed project, as it underscores the necessity of building an "effective privacy management system" that actively manages the consent lifecycle rather than treating it as a one-time transactional checkbox.

2. **European Union Agency for Cybersecurity (ENISA). (2014). Privacy and data protection by design – From engineering policy.**

ENISA's publication provides a technical blueprint for implementing "Privacy by Design" (PbD). It translates abstract legal policies into concrete software engineering practices. For the development of the Data Privacy Compliance System, this literature provides the structural guidelines for ensuring that data protection is hardcoded into the system's architecture (e.g., automated data logging and secure storage) rather than bolted on as an afterthought.

3. **Pandit, H. J., Debruyne, C., O'Sullivan, D., & Lewis, D. (2019). GConsent: A vocabulary for consent.** This conference paper introduces "GConsent," a standardized semantic vocabulary used to model and track user consent digitally. By establishing clear definitions for the state, purpose, and context of consent, the authors provide a framework for interoperability. This study informs the design of the proposed Consent Management Module, offering a theoretical model on how to accurately record, update, and retrieve the consent status of data subjects within a relational database.

4. **Tesfay, M. A., Hofmann, P., Nakamura, T., Kiyomoto, S., & Serna, J. (2018). PrivacyGuide: Towards an implementation of the EU GDPR on internet privacy policy evaluation.**
The authors of this paper developed "PrivacyGuide," a tool that evaluates internet privacy policies against the strict requirements of the EU GDPR. While the GDPR is a foreign regulation, the methodology used to automate policy evaluation and compliance tracking is directly applicable to this project. It highlights the importance of creating an Automated Reporting Dashboard that can parse system activities and map them against regulatory standards.

5. **Wachter, S. (2018). Normative challenges of identification in the Internet of Things: Privacy, profiling, discrimination, and the GDPR.** Wachter discusses the profound privacy risks associated with automated data collection and profiling, particularly in interconnected environments like the Internet of Things (IoT). This literature emphasizes the severe consequences of unchecked data processing. For the proposed system, it reinforces the need for a robust Data Processing Tracker that requires Data Protection

Officers (DPOs) to log and justify all internal data handling activities to prevent unauthorized profiling.

Local Literature

1. Bureau of Internal Revenue. (n.d.). Data privacy manual. Republic of the Philippines.

The BIR's manual serves as a prime example of how major Philippine government agencies interpret and operationalize RA 10173. It outlines strict internal protocols for handling taxpayer information. Reviewing this manual provides practical insights into the specific operational workflows and security measures expected of local organizations, which must be mirrored in the system's role-based access control (RBAC) features.

2. Department of Social Welfare and Development. (2023). Data privacy manual. Republic of the Philippines.

This manual details the DSWD's approach to protecting the highly sensitive personal data of marginalized sectors. It places a heavy emphasis on incident response and the handling of data subject requests. This literature validates the need for the proposed system to possess centralized, easily accessible logs that can quickly facilitate data retrieval or erasure if a data subject exercises their rights.

3. Learning Gate. (2025). Awareness of data privacy vis-à-vis data management practices at a state university. This recent local study evaluates the gap between theoretical data privacy awareness and actual, on-the-ground data management practices in an academic setting. The study frequently notes that human error and a lack of centralized tools contribute to poor data handling. This perfectly supports the problem statement of the proposed project, which identifies manual administration as a key driver of non-compliance and data breaches.

4. University of Southeastern Philippines. (2020). Data privacy manual.

USEP's privacy manual demonstrates the compliance strategy of a state university, focusing on the consent lifecycle of students and faculty—from enrollment to graduation. It provides a localized framework for how educational institutions categorize data (e.g.,

academic records vs. medical records), which is essential for designing the data categorization features within the project's Consent Management Module.

5. University of the Philippines Center for Integrative and Development Studies. (2025). Unlocking government data for evidenced-based and agile policymaking.

This paper explores the tension between utilizing data for agile policymaking and strictly adhering to data privacy laws. It argues that data must be unlocked securely to be useful. This informs the project's overarching philosophy: a compliance system should not just lock data away, but rather facilitate its secure, transparent, and legally compliant use through automated tracking and verifiable consent.

Related Systems

Foreign Systems

1. Ethyca Privacy Infrastructure

Ethyca provides a developer-focused infrastructure that automates data mapping and consent management across an organization's tech stack. By embedding privacy controls into the code layer, it eliminates manual tracking. Ethyca serves as a benchmark for the proposed project's "Privacy by Design" objective, showcasing how technical infrastructure can seamlessly translate legal requirements into automated developer tools.

2. OneTrust Privacy Management Software

OneTrust is a global enterprise leader in privacy management. It features comprehensive modules for cookie compliance, vendor risk management, and regulatory reporting. While OneTrust is vast and complex, analyzing its core functionality—specifically its dashboard that aggregates compliance metrics—provides a blueprint for designing the Automated Reporting Dashboard required for local DPOs.

3. Truyo Privacy Rights Platform

Truyo specializes in automating Subject Rights Requests (SRRs), allowing users to

independently request access to, or deletion of, their data without burdening administrative staff. This system highlights the importance of user-facing portals. It inspires the proposed project to not only build backend tools for the DPO but also to ensure the Consent Management Module interacts smoothly with the data subjects themselves.

Local Systems

1. National Privacy Commission (NPC) DPO ACE Portal

The DPO Accountability, Compliance, and Ethics (ACE) portal is an educational and certification platform by the NPC. While primarily a training system, it defines the standard curriculum and compliance expectations for DPOs in the Philippines.

Understanding the modules within the ACE portal helps ensure that the terminology and workflows built into the proposed project align perfectly with what the regulatory body teaches local DPOs.

- 2. National Privacy Commission Registration System (NPCRS)** The NPCRS is the official government portal where organizations must register their Data Processing Systems and DPOs. Because organizations are mandated to interface with this system, the proposed Data Privacy Compliance System must generate automated compliance reports and audit trails in a format that makes it easy for administrative staff to extract data and submit it to the NPCRS.

3. University of the Philippines Diliman Data Protection Office Portal

This institutional system serves the UP Diliman community, providing a centralized hub for privacy guidelines, incident reporting, and data processing requests. It functions as a localized, academic-specific privacy manager. Evaluating this system provides valuable insights into how a local organization customizes its interface to cater to its specific demographic (students, faculty, and researchers) while remaining compliant with national laws.

Relevant Laws

Applicable Law(s)

The primary legal framework governing this project is Republic Act No. 10173, known as the Data Privacy Act of 2012 (DPA). The DPA is the comprehensive privacy legislation of the Philippines, designed to protect the fundamental human right to privacy of communication while ensuring the free flow of information to promote innovation and growth.

Relevant Sections

The development of the Data Privacy Compliance System is anchored on several key sections of the DPA:

- **Section 11 (General Data Privacy Principles):** This section mandates that the processing of personal information must adhere to the principles of Transparency, Legitimate Purpose, and Proportionality. The system's Data Processing Tracker ensures these principles are met by logging the specific purpose and scope of all data handling activities.
- **Section 16 (Rights of the Data Subject):** This section outlines the rights of individuals, including the right to be informed, the right to object, and the right to erasure or blocking. The Consent Management Module is specifically designed to uphold these rights, allowing the organization to efficiently track and update user consent statuses securely.
- **Section 20 (Security of Personal Information):** This section requires personal information controllers to implement reasonable and appropriate organizational, physical, and technical security measures. The proposed system acts as a technical security measure by centralizing data logs, utilizing role-based access control (RBAC), and eliminating the vulnerabilities associated with scattered manual records.

Legal Basis of the Project

The legal basis for developing this system stems directly from the mandate of the Data Privacy Act of 2012, which holds organizations legally accountable for the personal data they process. Because non-compliance can result in severe penalties and reputational damage from the

National Privacy Commission (NPC), organizations possess a legal obligation to maintain accurate, verifiable records of consent and processing activities. By automating consent tracking and regulatory reporting, this project provides a necessary technological solution for organizations to fulfill their legal obligations under Philippine law efficiently and securely.

Chapter 3: Methodology

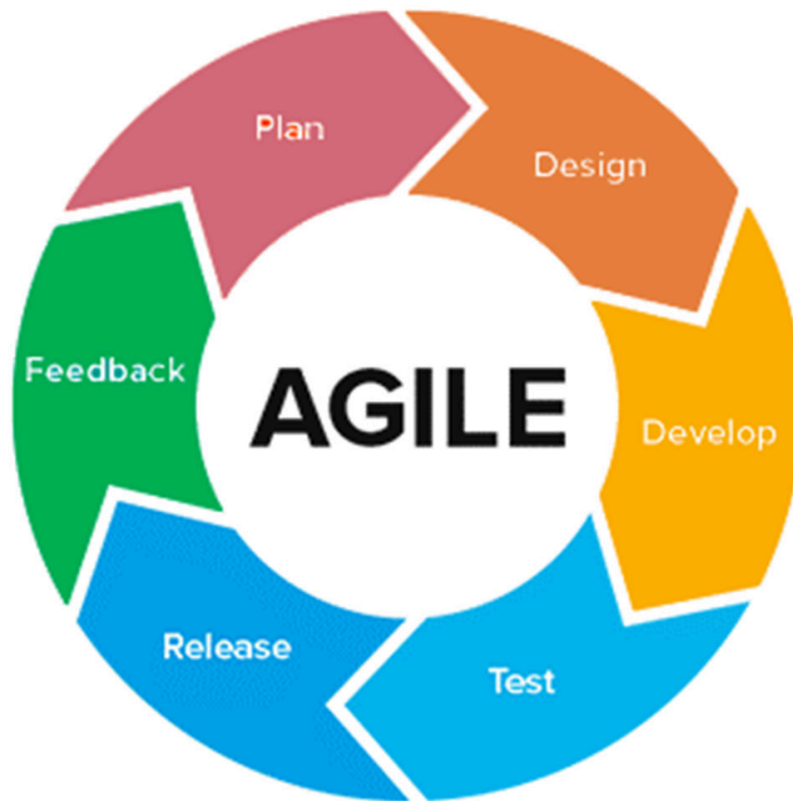
For the development of the Data Privacy Compliance System, the Agile methodology is the most highly suitable and effective choice. The Agile approach is centered on iterative development, continuous feedback, and adaptability, which perfectly aligns with the complex and sensitive nature of building a legally compliant, security-focused web platform. Because data privacy standards and the specific operational workflows of an organization can evolve, locking into a rigid, linear framework like Waterfall could result in a system that is outdated or non-compliant by the time it is finished. Agile ensures that the development process remains flexible and responsive to the intricate requirements of the Data Privacy Act of 2012 (RA 10173) and the National Privacy Commission (NPC).

The viability of Agile for this specific project lies heavily in its sprint-based, modular structure. The system is composed of several distinct and critical components, such as the Consent Management Module, the Data Processing Tracker, and the Automated Reporting Dashboard. Agile allows the development team to break these components down into manageable phases. For example, the team can focus an early sprint entirely on establishing secure user authentication and Role-Based Access Control (RBAC). Once that foundation is tested and approved, subsequent sprints can focus on building the consent tracking and audit logging

features. This incremental delivery ensures that each highly sensitive module is fully functional and secure before moving on to the next.

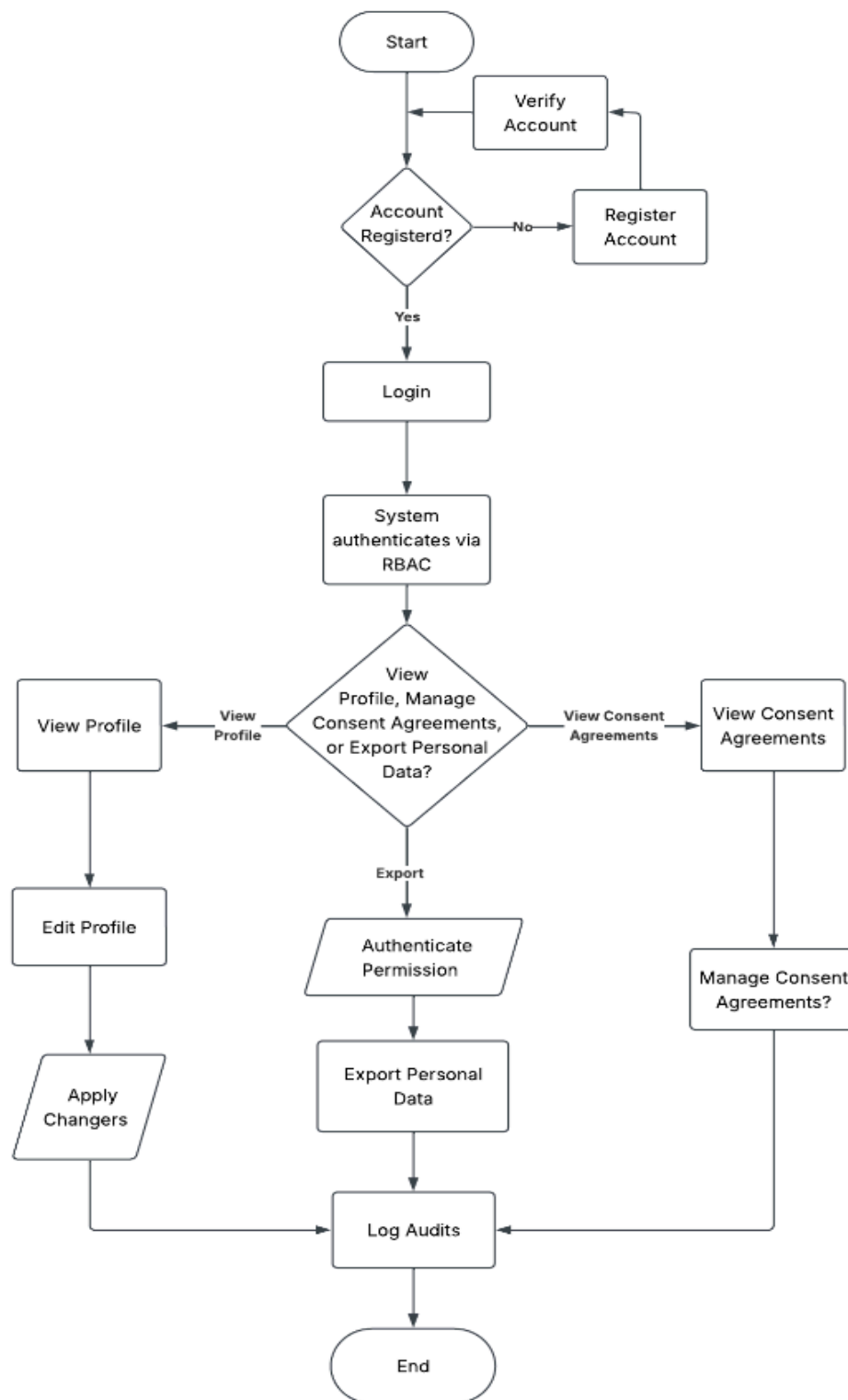
Furthermore, Agile facilitates continuous stakeholder involvement, which is vital when designing a system meant to alleviate the administrative burden on a Data Protection Officer (DPO). By releasing functional increments of the software at the end of each sprint, the DPO and administrative staff can test the system hands-on. If the Automated Reporting Dashboard does not align perfectly with the formats required by the NPC Registration System (NPCRS), the DPO can provide immediate feedback, allowing the developers to adjust the feature in the very next sprint. This continuous feedback loop guarantees that the final product will be a highly usable, practical tool rather than just a theoretical solution.

Finally, Agile perfectly complements the core philosophy of "Privacy by Design." Developing a system to handle personal information and prevent data breaches requires rigorous, ongoing testing. In an Agile environment, quality assurance and security testing occur continuously alongside development, rather than being pushed to the very end of the project lifecycle. This allows the team to consistently verify that data logging, secure storage, and incident reporting mechanisms operate flawlessly, drastically mitigating the risk of human error and ensuring the organization remains fully compliant with the stringent standards of Philippine privacy law.

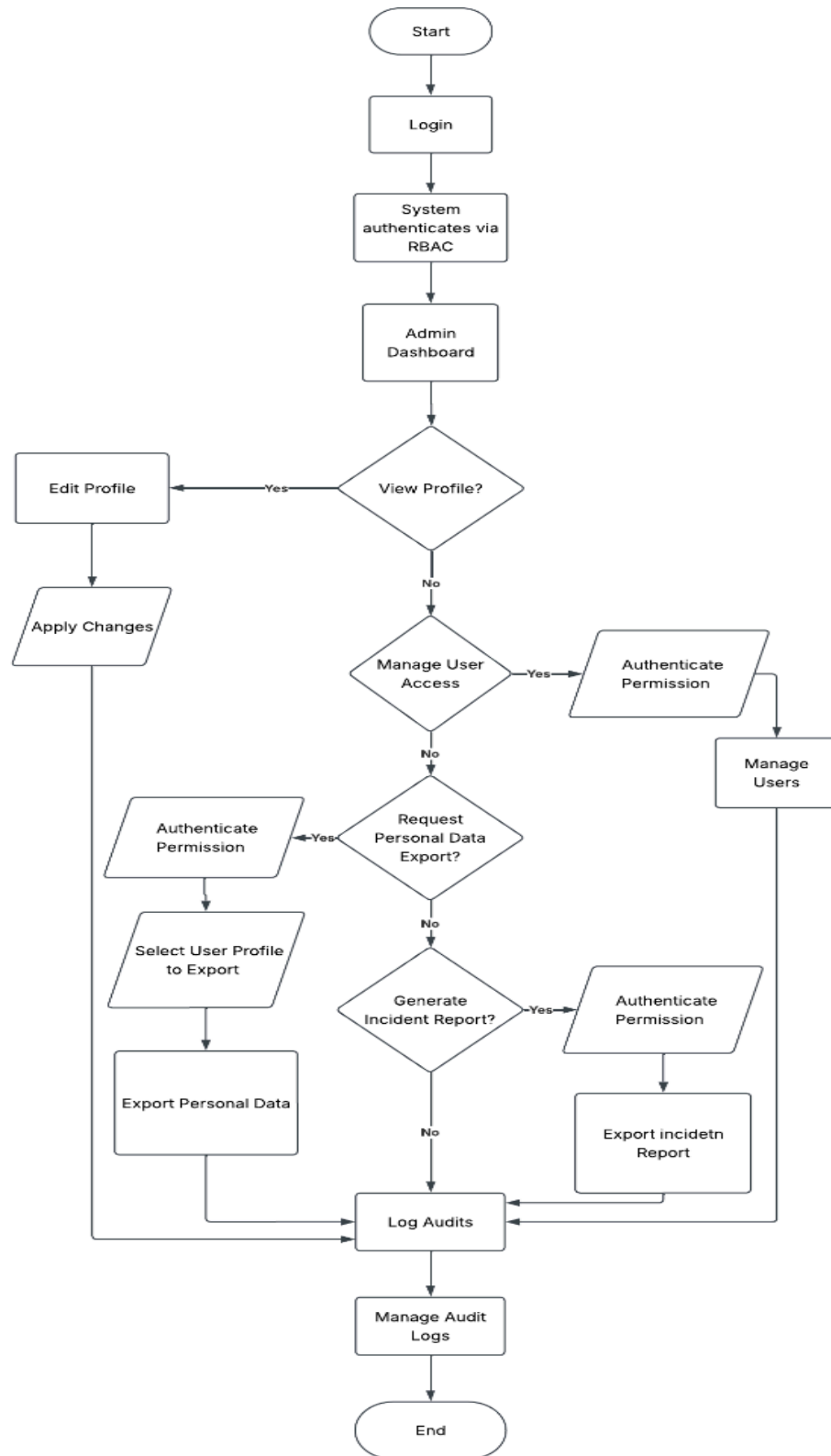


User Flow Diagrams

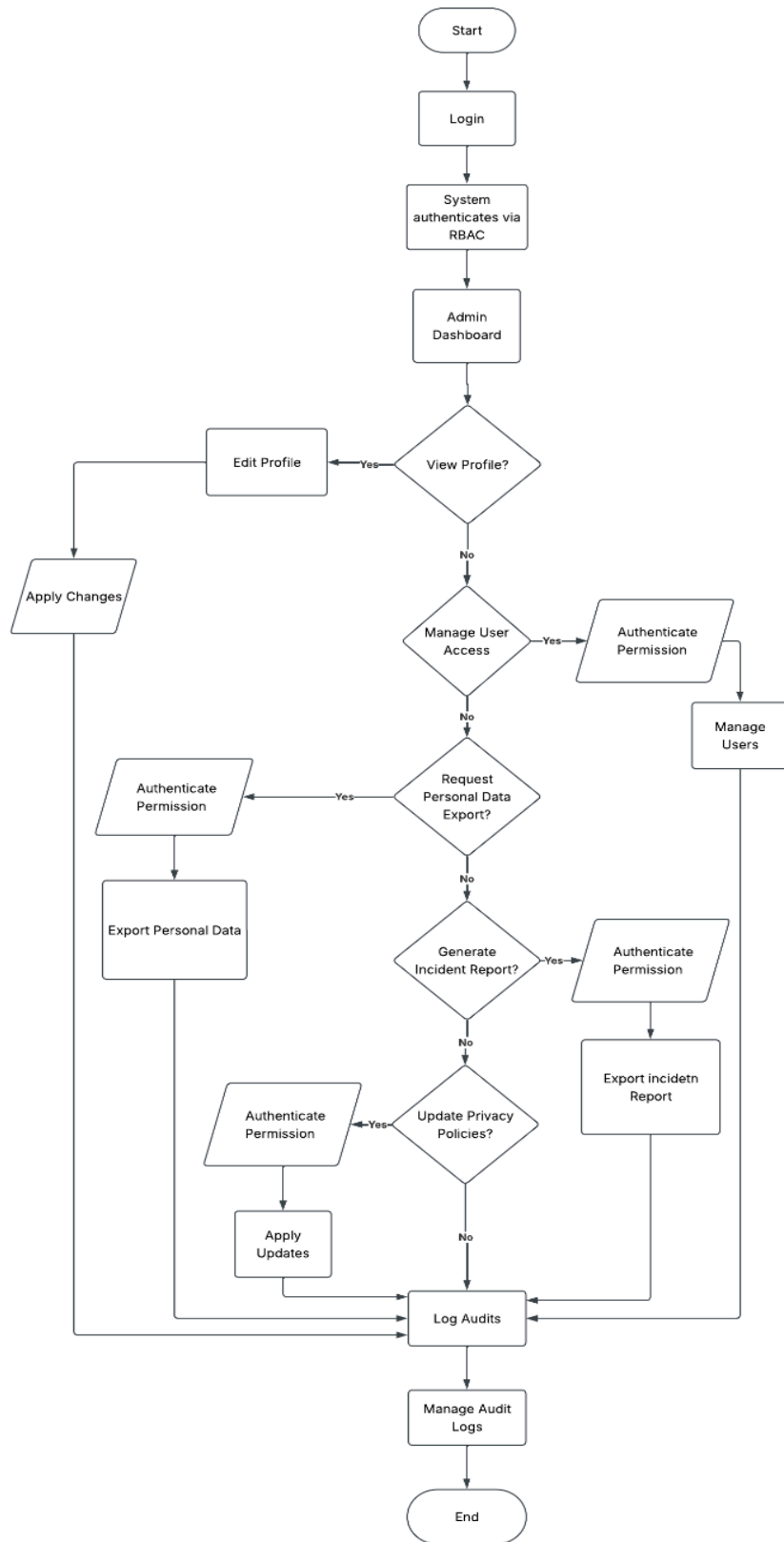
[Registered User Flow Diagram](#)



[Admin Flow Diagram](#)



Data Privacy Officer Flow Diagram

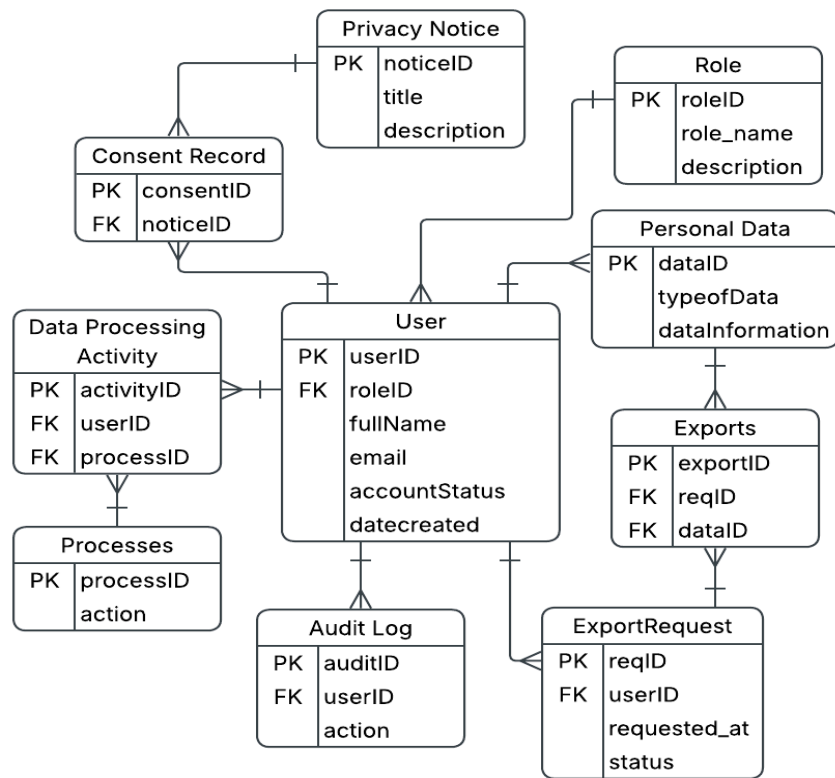


System Design

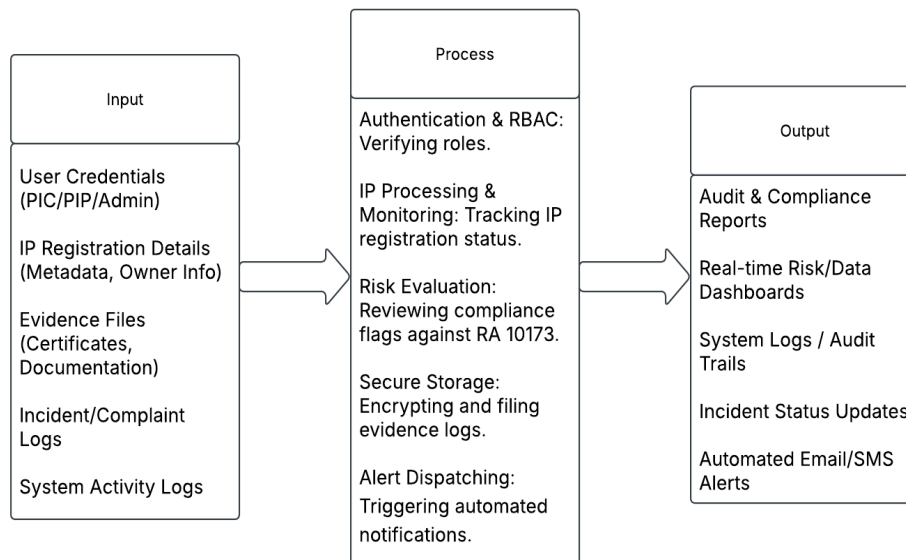
Use Case Diagram



Entity Relationship Diagram



IPO Diagram



Chapter 4: Results and Discussion

Chapter 5: Conclusions and Recommendations

Foreign Literature

1. **Betkier, M. (2019).** Moving beyond consent in data privacy law: An effective privacy management system for internet services [Doctoral thesis, Victoria University of Wellington]. Open Access Te Herenga Waka.
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<https://dpoace.privacy.gov.ph/>
2. **National Privacy Commission. (n.d.-b).** NPC registration system (NPCRS).
<https://npcregistration.privacy.gov.ph/>
3. **University of the Philippines Diliman. (n.d.).** Data protection office.
<https://privacy.upd.edu.ph/>

<https://privacy.gov.ph/data-privacy-act/>